

Definition of Drone

A drone is an unmanned aircraft that can be either operated remotely or autonomously using GPS, sensors, and AI-driven software.(MQ-9 Reaper,DIJ Agar series)



Definition of UAV

A UAV (Unmanned Aerial Vehicle) is an aircraft that flies without a human pilot inside it.

It is commonly called a drone



Difference between Drone and UAV

UAV	Drone
Stands for Unmanned Aerial Vehicle	Common name
Manily refers to the flying vechile itself	Refers to the whole Unmanned system
Uesd in Defence, Aerospace	Used in daily life



Components of Drones

- 1.Frame

The frame is the body skeleton Drone
- 2.Motors

Rotates the Propellers
- 3.Propellers

It cut the air and create lift
- 4.Electronic Speed Controller(ESU)

It controls the speed of the Motor
- 5.Flight Controller

The Brain of the drone process the data form the sensor, controls automatically
- 6.Battery

Supplies electrical power
- 7.GPS Module

Help in the navigation
- 8.Radio Tansmitter & Receiver

Used for the communication between Drone and Pilot
- 9.Sensors

Help stabilize and automate the drone(Gyroscope, Accelerometer,Barometer)
10. Camera

Used for the Video and Surveillance
- 11.Gimbal

A part that holds the camera
- 12.Landing Gear

Supports during landing



MAVlink

MAVLink is a communication language/protocol used for communication between a drone and devices like a ground station, flight controller, or computer.

Autopilot firmware

Autopilot firmware is the main software inside the flight controller that controls how the drone flies

PID Control

PID control is a method used to keep the drone stable and balanced automatically

MAIN COMPONENTS OF A DRONE



1. FRAME

The main body that holds all components together.



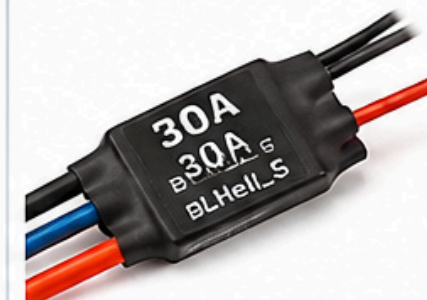
2. MOTORS

Rotate the propellers to generate thrust and lift.



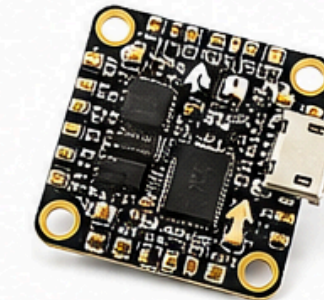
3. PROPELLERS

Spin and push air downwards to lift the drone.



4. ESC

Controls the speed of each motor.



5. FLIGHT CONTROLLER

The brain of the drone. Processes data from sensors and controls the flight.



6. BATTERY

Provides electrical power to all components.



7. GPS MODULE

Provides location and helps in navigation and auto functions.



8. RECEIVER

Receives signals from the remote controller and sends to flight controller.



9. CAMERA

Captures video or images during flight.



10. GIMBAL

Stabilizes the camera to get smooth and steady footage.



11. LANDING GEAR

Supports the drone during landing and take-off.



12. TELEMETRY MODULE

Sends flight data to the ground station (like battery, altitude, speed).



13. BUZZER

Gives sound alerts for warnings and status indications.



14. LED INDICATOR

Shows the status of the drone and arm/disarm state.



15. ANTENNA

Improves the range and strength of radio communication.